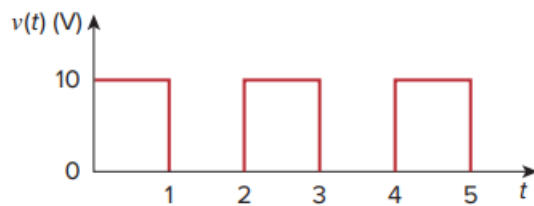


Problem

If the voltage waveform in Fig. 6.67 is applied across the terminals of a 5-H inductor, calculate the current through the inductor. Assume $i(0) = -1$ A.

Fig. 6.67:



Step-by-step solution

Step 1 of 2

$$i = \frac{1}{L} \int_0^t V dt + i(0)$$

$$i = \frac{1}{5} \int_0^t V(t) dt - 1$$

$$\text{For } 0 < t < 1, \quad i = \frac{10}{5} \int_0^t dt - 1$$

$$i = 2t - 1 \text{ A}$$

$$\text{For } 1 < t < 2, \quad i = 0 + i(1)$$

$$i = 1 \text{ A}$$

$$\text{For } 2 < t < 3, \quad i = \frac{1}{5} \int_2^t 10 dt + i(2)$$

$$i = 2t \Big|_2^t + 1$$

$$i = 2t - 3 \text{ A}$$

$$\text{For } 3 < t < 4, \quad i = 0 + i(3)$$

$$i = 3 \text{ A}$$

$$\text{For } 4 < t < 5, \quad i = \frac{1}{5} \int_4^t 10 dt + i(4)$$

$$i = 2t \Big|_4^t + 3$$

$$i = 2t - 5 \text{ A}$$

Step 2 of 2

Thus,

$$i(t) = \begin{cases} 2t - 1 \text{ A}, & 0 < t < 1 \\ 1 \text{ A}, & 1 < t < 2 \\ 2t - 3 \text{ A}, & 2 < t < 3 \\ 3 \text{ A}, & 3 < t < 4 \\ 2t - 5 \text{ A}, & 4 < t < 5 \end{cases}$$